9.3 Telescoping Sums Worksheet

Name: _____ Date: _____

Telescoping Sums

Find a closed form for the nth partial sum s_n (use partial fractions or log rules), then determine whether the series converges by computing $\lim_{n\to+\infty} s_n$. If it converges, find its sum.

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1. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^2+3k+2} = \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \cdots + \frac{1}{k^2+3k+2} + \cdots$

☐ converges, sum = _____ ☐ diverges

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2. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{(2k-1)(2k+1)} = \frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} + \cdots + \frac{1}{(2k-1)(2k+1)} + \cdots$

☐ converges, sum = _____ ☐ diverges

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3. $\displaystyle\sum_{k=1}^{\infty} \ln\left(\frac{k}{k+1}\right) = \ln\frac{1}{2} + \ln\frac{2}{3} + \ln\frac{3}{4} + \cdots + \ln\left(\frac{k}{k+1}\right) + \cdots$

☐ converges, sum = _____ ☐ diverges

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4. $\displaystyle\sum_{k=1}^{\infty} \frac{2}{k(k+2)} = \frac{2}{1\cdot 3} + \frac{2}{2\cdot 4} + \frac{2}{3\cdot 5} + \cdots + \frac{2}{k(k+2)} + \cdots$

☐ converges, sum = _____ ☐ diverges

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5. $\displaystyle\sum_{k=1}^{\infty} \frac{4}{k^2+4k+3} = \frac{4}{2\cdot 4} + \frac{4}{3\cdot 5} + \frac{4}{4\cdot 6} + \cdots + \frac{4}{k^2+4k+3} + \cdots$

☐ converges, sum = _____ ☐ diverges

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6. $\displaystyle\frac{1}{4\cdot 5} + \frac{1}{5\cdot 6} + \frac{1}{6\cdot 7} + \cdots + \frac{1}{(k+3)(k+4)} + \cdots$

☐ converges, sum = _____ ☐ diverges

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7. $\displaystyle\frac{1}{1\cdot 4} + \frac{1}{4\cdot 7} + \frac{1}{7\cdot 10} + \cdots + \frac{1}{(3k-2)(3k+1)} + \cdots$

☐ converges, sum = _____ ☐ diverges